

Fumiyuki Kato

PhD student at Kyoto University
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EDUCATION

Doctor of Informatics at Kyoto University

April 2021 - present

Focus on Privacy and Security of data engineering, especially on Differential Privacy, Federated Learning and Trusted Execution environment. (supervised by Dr. Yang Cao and Prof. Masatoshi Yoshikawa)

Master of Informatics at Kyoto University

April 2019 - March 2021

Master Thesis: PCT-TEE: Trajectory-based Private Contact Tracing System with Trusted Execution Environment. (supervised by Dr. Yang Cao and Prof. Masatoshi Yoshikawa)

Bachelor of Informatics at Kyoto University

April 2015 - March 2019

Bachelor Thesis: Future Prediction based on Web News Articles with Publication Source and Time. (supervised by Prof. Makoto P. Kato and Prof. Masatoshi Yoshikawa)

PUBLICATIONS

- [1]. **Fumiyuki,Kato.**, Tsubasa, Takahashi., Shun, Takagi., Yang, Cao., Seng Pei, Liew., & Masatoshi, Yoshikawa. HDPView: Differentially Private Materialized View for Exploring High Dimensional Relational Data. In International Conference on Very Large Databases (VLDB). 2022.
- [2]. Seng Pei, Liew., Tsubasa, Takahashi., Shun, Takagi., **Fumiyuki,Kato.**, Yang, Cao., & Masatoshi, Yoshikawa. Network Shuffling: Privacy Amplification via Random Walks. In International Conference on Management of Data (SIGMOD). 2022.
- [3]. **Fumiyuki,Kato.**, Yang, Cao., & Masatoshi, Yoshikawa. PCT-TEE: Trajectory-based Private Contact Tracing System with Trusted Execution Environment. In ACM Transactions on Spatial Algorithms and Systems (TSAS). 2021.
- [4]. **Fumiyuki,Kato.**, Yang, Cao., & Masatoshi, Yoshikawa. Preventing Manipulation Attack in Local Differential Privacy using Verifiable Randomization Mechanism. In 35th Annual IFIP WG 11.3 Conference, DBSec (DBSec). 2021.
- [5]. **Fumiyuki,Kato.**, Yang, Cao., & Masatoshi, Yoshikawa. Secure and Efficient Trajectory-Based Contact Tracing using Trusted Hardware. In 7th International Workshop on Privacy and Security of Big Data at IEEE BigData 2020.
- [6]. Zheng, Wen., Keping, Yu., Xin, Qi., Toshio, Sato., Yutaka, Katsuyama.,Takuro, Sato., Wataru, Kameyama., **Fumiyuki,Kato.**, Yang, Cao., Masatoshi, Yoshikawa., Min, Luo., & Jun, Hashimoto. Blockchain-empowered Contact Tracing for COVID-19 Using Crypto-spatiotemporal Information. In 2020 IEEE International Conference on E-health Networking, Application & Services (HEALTHCOM).

AWARDS

- (Feb.2022) Excellent Research Award at Kyoto University's 16th ICT Innovation
- (March.2021) Student Presentation Award (DEIM 2021)
- (March.2021) Master thesis award of social informatics department of Kyoto University
- (March.2020) Student Presentation Award (DEIM 2020)

FUNDINGS

- (April.2021 - March.2024) Kyoto University AI/Data science Fellowship. (2,100,000 yen / year)
- (March.2021) Exemption from repayment of scholarships of JASSO due to outstanding achievements. (half)

SKILLS AND QUALIFICATIONS

Programming Languages

<i>Advanced</i>	Python
<i>Basic</i>	C++, Rust, Perl, Java, JS, TS, Web applications, Pytorch

Languages

<i>Native tongue</i>	Japanese
<i>Basics</i>	English

WORK EXPERIENCE

Research intern at LINE Corporation. *August 2020 - March 2021*

Research privacy enhanced technologies. The research outcome is published at VLDB'22.

Web application engineer (part-time) at Hatena Inc. *August 2020 - March 2021*

Developing backend and frontend of large scale web applications mainly using Perl and Javascript.

Software engineer (intern) at Recruit Co.,Ltd. *October 2019 - November 2019*

Developing backend and frontend large scale web applications mainly using Spring Boot and Java.

Data scientist (intern) at Wantedly Inc. *September 2019*

Working on proof of concept of a recommendation of a web application using Python and Ruby and BigQuery.

ML engineer (intern) at M3 Inc. *August 2019*

Developing backend infrastructure of recommendation service of web applications mainly using Scala and Kubernetes.

Software engineer (part-time) at MediaMaxJapan Inc. *July 2018 - March 2019*

Developing web applications mainly using Kotlin and Typescript. In particular, I was able to experience development that strictly followed DDD.

Software engineer (part-time) at KiizanKiizan Inc.

Feburary 2018 - May 2018

Developing web applications mainly using Ruby.